NLP PRACTICAL EXERCISE - 04

Aim:
To implement Chunking by defining grammar rules to extract meaningful phrases such as Noun Phrases (NP) and Verb Phrases (VP)

Procedure :

Step 1 — Open Google Colab
Step 2 — Import NLTK
CODE - import nltk

Step 3 — Download required packages
CODE -
```
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
```

Step 4 — Get user input
CODE -
```
text = input("Enter a sentence: ")
print("Input:", text)
```

Step 5 — Tokenization
CODE -
```
tokens = nltk.word_tokenize(text)
print("Tokens:")
print(tokens)
```

Step 6 — POS Tagging
CODE -
```
pos_tags = nltk.pos_tag(tokens)
print("POS Tags:")
print(pos_tags)
```

Step 7 — Define Chunking Grammar

CODE -

```
grammar = r"""
   NP: {<DT>?<JJ>*<NN.*>}
   VP: {<VB.*><NP|PP>*}
"""
```

MEANING : NP = Noun Phrase VP = Verb Phrase

Step 8 — Create Chunk Parser

CODE -

```
chunk_parser = nltk.RegexpParser(grammar)
```

Step 9 — Perform Chunking

CODE -

```
tree = chunk_parser.parse(pos_tags)
print(tree)
```

Step 10 — Display Meaningful Phrases

CODE -

```
print("Meaningful Phrases:")

for subtree in tree.subtrees():
   if subtree.label() in ['NP', 'VP']:
      phrase = " ".join(word for word, tag in subtree.leaves())
      print(subtree.label(), ":", phrase)
```

Result:

Thus, meaningful phrases such as Noun Phrases (NP) and Verb Phrases (VP) were successfully extracted using NLTK chunking.

EXPECTED OUTPUT : (SAMPLE)



```
[3] ✓ 9s
text = input("Enter a sentence: ")
print("Input:", text)

Enter a sentence: The small dog chased the cat.
Input: The small dog chased the cat.

[4] ✓ 0s
tokens = nltk.word_tokenize(text)

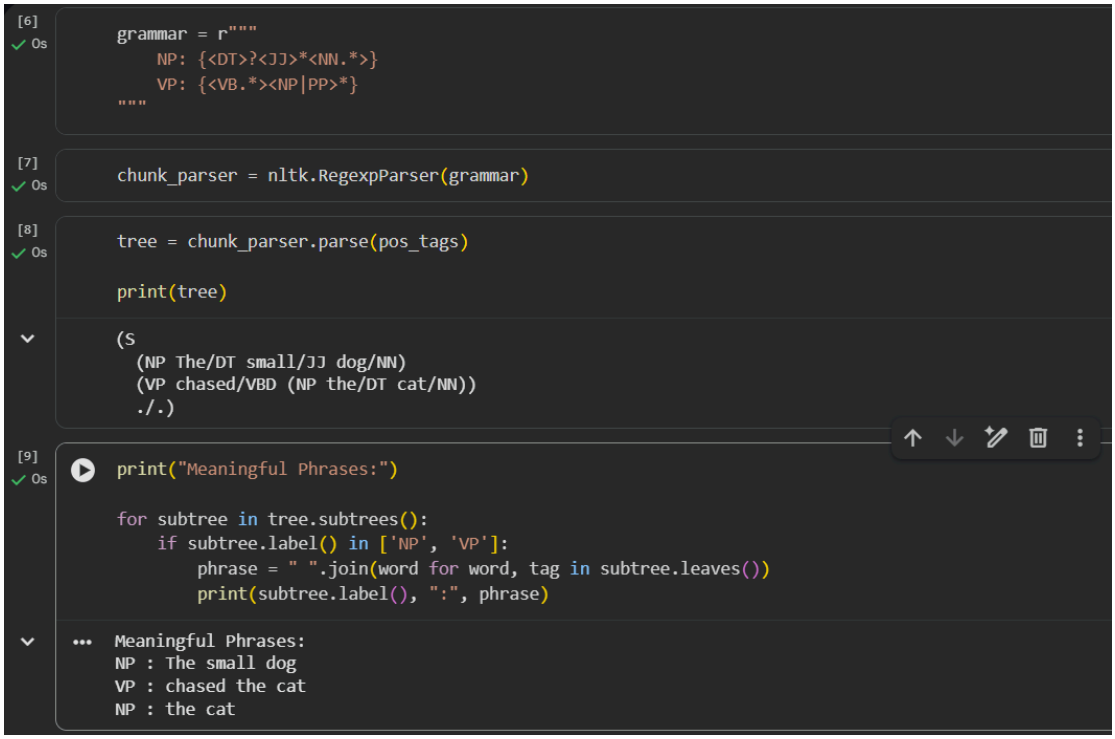
print("Tokens:")
print(tokens)

Tokens:
['The', 'small', 'dog', 'chased', 'the', 'cat', '.']

[5] ✓ 0s
pos_tags = nltk.pos_tag(tokens)

print("POS Tags:")
print(pos_tags)

POS Tags:
[('The', 'DT'), ('small', 'JJ'), ('dog', 'NN'), ('chased', 'VBD'), ('the', 'DT'), ('cat', 'NN'), ('
```



```
[6] ✓ 0s
grammar = r"""
    NP: {<DT>?<JJ>*<NN.*>}
    VP: {<VB.*><NP|PP>*}
"""

[7] ✓ 0s
chunk_parser = nltk.RegexpParser(grammar)

[8] ✓ 0s
tree = chunk_parser.parse(pos_tags)

print(tree)

(S
  (NP The/DT small/JJ dog/NN)
  (VP chased/VBD (NP the/DT cat/NN))
  ./.)

[9] ✓ 0s
print("Meaningful Phrases:")

for subtree in tree.subtrees():
    if subtree.label() in ['NP', 'VP']:
        phrase = " ".join(word for word, tag in subtree.leaves())
        print(subtree.label(), ":", phrase)

Meaningful Phrases:
NP : The small dog
VP : chased the cat
NP : the cat
```